

Lecture 12

Scattering without diffraction: the open Toda chain

Objectives. Course map: the bridge from P1 to P3, classically. [Lecture 10](#) and [Lecture 11](#) found two-body phases by guessing wavefunctions, and a quantum many-body state built from them alone. This lecture shows why that works, in classical mechanics, where it can be watched. Open the Toda ring into a chain: its particles fly apart, their final momenta are the eigenvalues of its Lax matrix, and the scattering of two particles or three or N only permutes the momenta and shifts each line by a sum of two-body shifts. The two-body shift is the classical counterpart of [Lecture 11](#)'s phase, and the agreement of the two ways of composing three two-body scatterings is what [Lecture 13](#) turns into the Yang–Baxter equation.

12.1 The question

[Lecture 0](#) showed three Toda particles leaving a collision with the momenta they came in with, rearranged, and promised the reason. So the question is

what does integrability do to scattering: when particles meet on a line and fly apart, what is conserved?

The answer: the set of momenta, which are the eigenvalues of the Lax matrix ([Theorem 12.2](#)), and the scattering map is a product of two-body maps, one per pair; for three particles the two ways of composing them agree ([Figure 12.6](#)). The worked example is two particles, in closed form ([Proposition 12.1](#)).

12.2 FPUT, and the Toda potential

[Lecture 0](#) told the story of the Fermi–Pasta–Ulam–Tsingou chain, whose spring energy ([0.2](#)) is weakly nonlinear and whose energy came back to the lowest mode instead of spreading (a reading at the end recalls it).

In 1967 Toda looked for a spring law for which the chain has *exact* periodic nonlinear waves, trying elliptic functions as the nonlinear analogue of sines, and found the exponential (Toda, §2.1). With the constants set to one and $r_n = q_{n+1} - q_n$,

$$V_{\text{Toda}}(r) = e^{-r} + r - 1 = \frac{r^2}{2} - \frac{r^3}{6} + \frac{r^4}{24} - \cdots, \quad (12.1)$$

which to cubic order is the FPUT spring ([0.2](#)), with $\alpha = -\frac{1}{2}$ ([Figure 12.1](#)). Compressed ($r < 0$) it is a steep exponential wall; stretched, it softens to the constant force of the linear term.

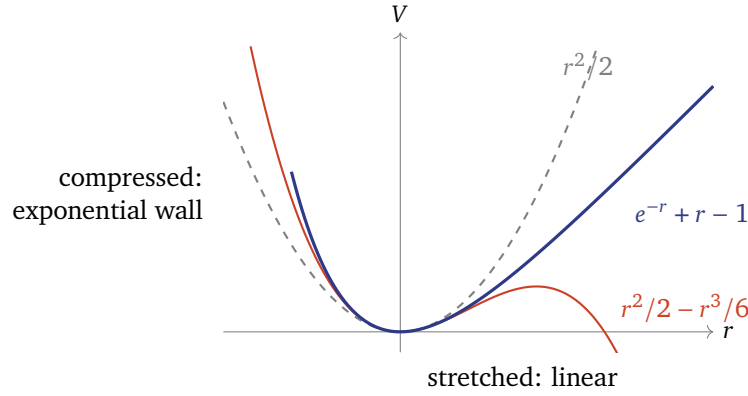


Figure 12.1: The Toda spring (12.1) (indigo), its harmonic approximation (dashed) and its cubic truncation, which is the FPUT spring with $\alpha = -\frac{1}{2}$ (red). All three agree to second order; the Toda spring is the one whose chain is exactly solvable.

For a *ring* of springs the linear terms add up to $\sum_n r_n =$ the circumference, a constant, and can be dropped; the result is the Toda ring of [Lecture 9](#), whose Poincaré section is [Figure 7.5](#). For an *open* chain nothing holds the ends, and we drop the linear terms by fiat.¹ What is left is the open chain of [Lecture 8](#),

$$H = \frac{1}{2} \sum_{n=1}^N p_n^2 + \sum_{n=1}^{N-1} e^{q_n - q_{n+1}}, \quad (12.2)$$

N unit masses on a line, each repelled by its neighbours ([Figure 12.2](#)). The repulsion is felt only when particle n comes close to, or overtakes, particle $n + 1$. So the open chain has no FPUT oscillations: its particles fly apart. It is a nonlinear many-body system simple enough to take apart completely.

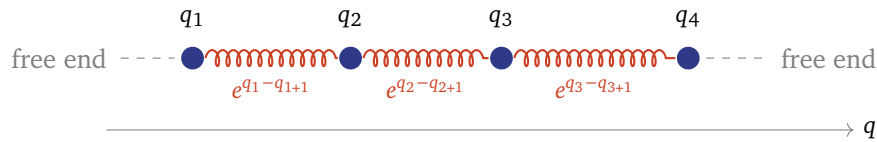


Figure 12.2: The open Toda chain, $N = 4$. Neighbours interact through $e^{q_n - q_{n+1}}$, which is small while particle n is well behind particle $n + 1$ and grows exponentially as it catches up. The end particles have one neighbour each.

It is [Lecture 9](#)'s Toda ring with its closing spring removed, and its Lax matrix is the tridiagonal one of [Proposition 8.3](#).

12.3 Two particles in closed form

Take $N = 2$: $H = \frac{1}{2}(p_1^2 + p_2^2) + e^{q_1 - q_2}$. The mean momentum $\bar{p} = \frac{1}{2}(p_1 + p_2)$ is conserved, and the centre of mass $\frac{1}{2}(q_1 + q_2)$ moves at the constant speed $\bar{p}$. The relative coordinate $x = q_1 - q_2$ obeys

$$\dot{x} = p_1 - p_2, \quad \ddot{x} = \dot{p}_1 - \dot{p}_2 = -2e^x,$$

¹Toda, §3.10, following Moser: the end particles are sent to infinity, where the linear terms are an infinite constant.

and since $\frac{1}{2}(p_1^2 + p_2^2) = \bar{p}^2 + \frac{1}{4}(p_1 - p_2)^2$, the energy of the relative motion is

$$H - \bar{p}^2 = \frac{\dot{x}^2}{4} + e^x =: \frac{v^2}{4}, \quad v > 0. \quad (12.3)$$

It is positive because e^x is, and it fixes v . Particle 1 starts behind particle 2 ($x \rightarrow -\infty$, where the potential is negligible) and approaches at relative speed v ; the wall e^x stops it at $e^x = v^2/4$ and sends it back.

Proposition 12.1 (The two-particle chain). *Every solution with $N = 2$ is, for some t_0 and a choice of origin,*

$$x(t) = \ln \frac{v^2}{4} - 2 \ln \cosh \frac{v(t-t_0)}{2}, \quad q_{1,2}(t) = \bar{p}t \pm \frac{x(t)}{2}. \quad (12.4)$$

Its properties:

- (a) $\dot{x} = -v \tanh \frac{v(t-t_0)}{2}$ runs from $+v$ to $-v$: the momenta $p_{1,2} = \bar{p} \pm \frac{1}{2}\dot{x}$ are exchanged, from $\bar{p} + \frac{v}{2}$, $\bar{p} - \frac{v}{2}$ to $\bar{p} - \frac{v}{2}$, $\bar{p} + \frac{v}{2}$.
- (b) At every time the eigenvalues of L are $\lambda_{1,2} = \bar{p} \mp \frac{v}{2}$: the asymptotic momenta, and $v = \lambda_2 - \lambda_1$.
- (c) With $t_0 = 0$, as $t \rightarrow -\infty$ and $t \rightarrow +\infty$,

$$q_1 \simeq \lambda_2 t + \ln v \quad (\text{in}), \quad q_2 \simeq \lambda_2 t - \ln v \quad (\text{out}),$$

and the same with λ_1 and the signs of $\ln v$ reversed for the other pair. The straight line carrying the momentum λ_2 is displaced by $-2 \ln v$, the one carrying λ_1 by $+2 \ln v$.

Proof. Differentiating (12.4), $\ddot{x} = -\frac{v^2}{2} \operatorname{sech}^2 \frac{v(t-t_0)}{2} = -2e^x$, and $\frac{1}{4}\dot{x}^2 + e^x = \frac{v^2}{4}(\tanh^2 + \operatorname{sech}^2) = \frac{v^2}{4}$. Every solution has exactly one turning point: $\ddot{x} < 0$, and $\dot{x}$ cannot stay positive, since then x increases and $\ddot{x} \leq -2e^{x(0)}$; likewise backwards in time. A solution is fixed by its energy and the time t_0 of its turning point, where $x = \ln(v^2/4)$ and $\dot{x} = 0$, and (12.4) has both free. (a) follows. For (b), $\operatorname{tr} L = 2\bar{p}$ and $\det L = p_1 p_2 - e^x = \bar{p}^2 - \frac{1}{4}\dot{x}^2 - e^x = \bar{p}^2 - \frac{1}{4}v^2$ by (12.3), so the characteristic polynomial is $(\eta - \lambda_1)(\eta - \lambda_2)$ for all t . For (c), use $\ln \cosh u = |u| - \ln 2 + o(1)$: then $x = \ln v^2 - v|t| + o(1)$, and $q_1 = \bar{p}t + \frac{x}{2} \simeq (\bar{p} + \frac{v}{2})t + \ln v$ as $t \rightarrow -\infty$, $q_2 = \bar{p}t - \frac{x}{2} \simeq (\bar{p} - \frac{v}{2})t - \ln v$ as $t \rightarrow +\infty$. ■

Read Figure 12.3 as a scattering experiment. In the far past there are two free particles with momenta $\lambda_2 > \lambda_1$, the faster one behind; in the far future, two free particles with the *same two momenta*, the faster one in front. Following the momentum rather than the particle, it is as if they had passed through each other, except that each straight line is displaced: the fast one by $-2 \ln v$, a time delay $2 \ln v/\lambda_2$ when $\lambda_2 > 0$. This displacement is the classical counterpart of the two-body phase shift of the delta gas in Lecture 11. For $v < 1$ its sign reverses: slow particles turn round while still far apart, and the momentum is handed forward across the gap.

12.4 General N : elastic, non-diffractive scattering

For N particles the general solution has determinant formulas (Moser; Babelon–Bernard–Talon, §4.8), which we omit: the asymptotics are all we need.

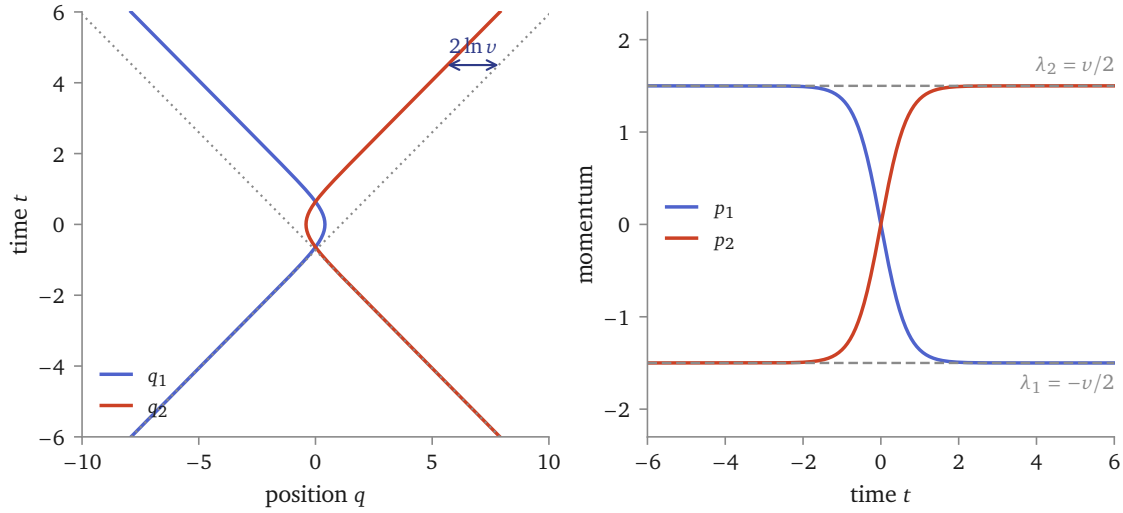


Figure 12.3: Two Toda particles, centre-of-mass frame, $v = 3$. Left: world lines (time upward). The dotted lines continue the *incoming* straight lines; after the collision each momentum travels on the other particle, on a line parallel to its incoming one and displaced by $2 \ln v$. Right: the momenta are exchanged, and the eigenvalues of L (dashed) are the asymptotic momenta $\pm v/2$ at every time (scripts/figures/toda_two_body.py, which checks (12.4) against a direct integration and the spectrum at every step).

Theorem 12.2 (Moser · cited). For every solution of the open Toda chain, $q_{n+1}(t) - q_n(t) \rightarrow +\infty$ as $t \rightarrow \pm\infty$, and each particle moves asymptotically on a straight line: $p_n(t) \rightarrow p_n^\pm$ and $q_n(t) = p_n^\pm t + \beta_n^\pm + o(1)$. With $\lambda_1 < \dots < \lambda_N$ the eigenvalues of L ,

$$p_n^+ = \lambda_n, \quad p_n^- = \lambda_{N+1-n}. \quad (12.5)$$

The line carrying λ_k arrives on particle $N+1-k$ and leaves on particle k , displaced by

$$D_k := \beta_k^+ - \beta_{N+1-k}^- = \sum_{j>k} 2 \ln(\lambda_j - \lambda_k) - \sum_{j<k} 2 \ln(\lambda_k - \lambda_j). \quad (12.6)$$

Given the separation, (12.5) is one line. If $q_{n+1} - q_n \rightarrow \infty$ then every $a_n \rightarrow 0$ and $L(t) \rightarrow \text{diag}(p^\pm)$. Eigenvalues depend continuously on the matrix and do not change in time, so $\{p_n^\pm\}$ is the spectrum $\{\lambda_k\}$. The separation fixes the order: $q_{n+1} - q_n \rightarrow +\infty$ as $t \rightarrow +\infty$ needs $p_{n+1}^+ \geq p_n^+$, so the outgoing momenta increase along the chain, and as $t \rightarrow -\infty$ they decrease. This is what the eigenvalues of L mean, which is what Figure 8.3 was hinting at: they are the momenta the particles have when they are free. The separation and (12.6) are Moser's (1975, pp. 480–481, in his normalisation; Toda, §3.10, gives (12.5) with L halved); scripts/check-l9-algebra.py integrates $N = 3, 4$ in ours and finds both to within 10^{-8} .

Two words describe this (Figure 12.4). The scattering is *elastic*: the set of outgoing momenta equals the set of incoming ones, so each particle's kinetic energy is handed on intact rather than shared. And it is *non-diffractive*: the outgoing momenta are not merely constrained by energy and momentum, which for $N \geq 3$ would allow a continuum of outcomes; they are exactly the incoming ones, permuted. N conserved eigenvalues leave room for nothing else.

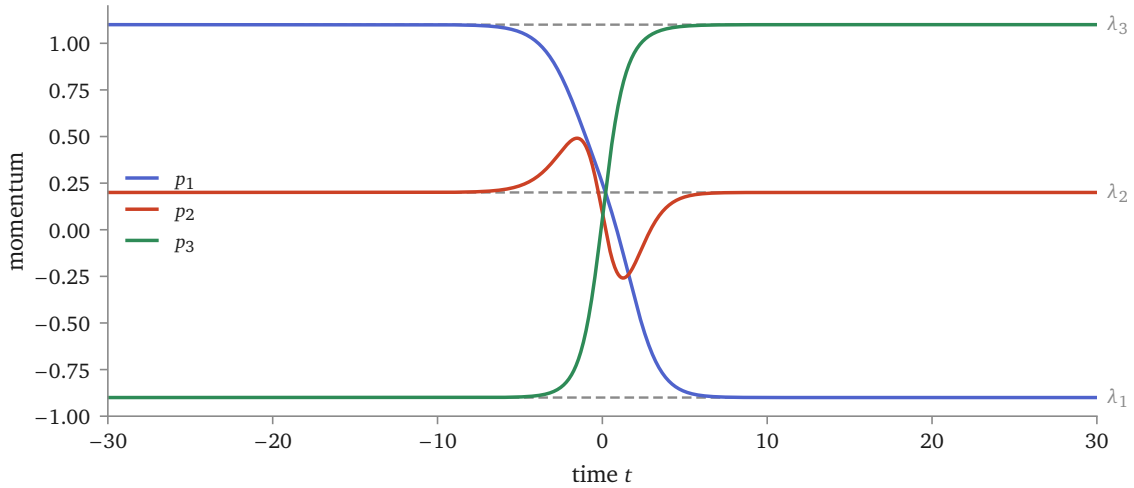


Figure 12.4: Three particles scatter. Their momenta (solid) start on the eigenvalues of L (dashed) in decreasing order and end on them in increasing order; in between, the eigenvalues do not move (`scripts/figures/toda_scattering.py`, which asserts the spectrum constant to 10^{-9} and both limits).

Three particles: collisions in either order

Look at (12.6) term by term. The line carrying λ_k picks up $+2 \ln(\lambda_j - \lambda_k)$ from every faster momentum and $-2 \ln(\lambda_k - \lambda_j)$ from every slower one: exactly the two-body displacements of Proposition 12.1, one per pair, added. The total depends on the eigenvalues only. In particular it is the same whichever pair the incoming data bring together first.

Figure 12.5 makes it concrete. With incoming momenta 3, 0, -3 on particles 1, 2, 3, started well apart, the run looks like three pairwise exchanges of momentum, which reverse the order, in the order (12), (23), (12) or (23), (12), (23). The forces never switch off, so the exchanges are only nearly separate; the net result, (12.6), is exact. The two runs start from different incoming lines and look different throughout, but each outgoing line is displaced from its own incoming line by the same amount in both. Written as permutations of the three momentum lines, $s_1 s_2 s_1 = s_2 s_1 s_2$ (Figure 12.6), and the displacements respect it.

This is the concrete content of the phrase *factorised scattering*: the N -body scattering map is a product of two-body maps, one per pair, taken along a route of crossings of adjacent lines from the incoming order of the lines to the outgoing one, and every such route gives the same map. The general theorem, that the N -body scattering map of the open chain factorises this way, is Moser's · *cited*. It is a statement about the asymptotic motion: the particles scatter as if they met in pairs, although for $N \geq 3$ more than two of them can interact at once. What we have seen is the two-body map in closed form and the three-body composition checked. Its quantum form is Lecture 11's and Lecture 13's: the two-body S-matrix has a phase whose derivative gives the packet displacement, and the braid relation becomes the Yang–Baxter equation.

12.5 What it buys

- *The meaning of the Lax eigenvalues*: the momenta the particles carry when they are free, which is what Lecture 0's collision figure showed.
- *Scattering without diffraction*, the second face of integrability: N -body scattering only

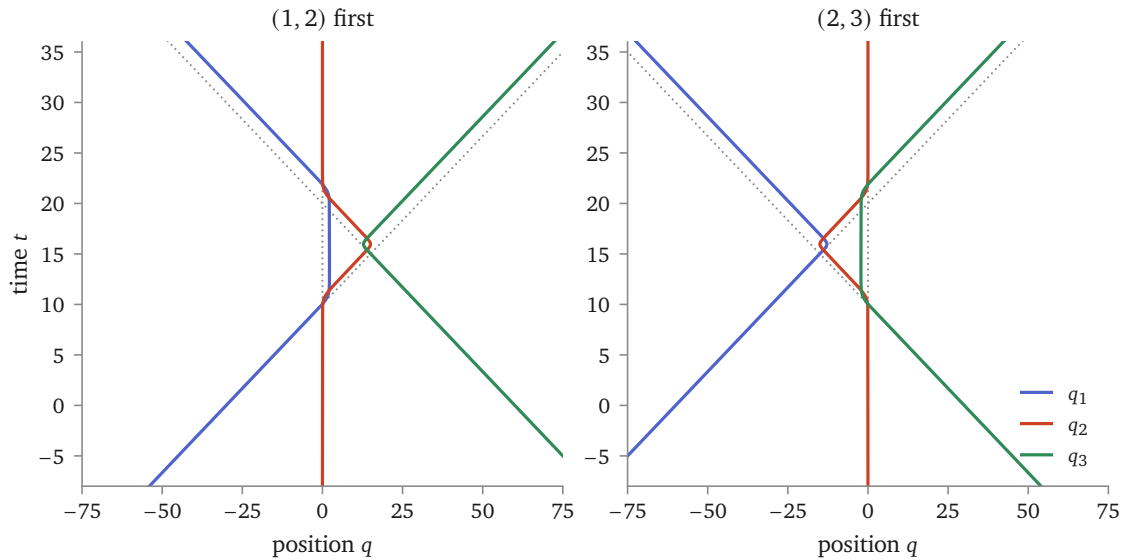


Figure 12.5: Three particles with the same incoming momenta 3, 0, -3 and different starting positions. Left, particles 1 and 2 meet first; right, 2 and 3 do. With the particles this far apart, each run looks like three two-body collisions, in different orders. The dotted lines continue the incoming lines; the net displacements are the same in both runs, $D = (2 \ln 18, 0, -2 \ln 18)$, as (12.6) says (scripts/figures/toda_three_body.py, which asserts it; the check script also runs a genuine three-body collision and gets the same).

permutes the momenta and shifts each line by a sum of two-body shifts.

- *The classical shadow of Part II's key structures:* the two-body shift against Lecture 11's phase, and the braid relation, which with matrices at the crossings is Lecture 13's Yang–Baxter equation.

12.6 Caveats, in one place

- *Status.* Proposition 12.1 is proved. Moser's theorem, the separation of the particles and the N -body displacements (12.6), is cited; given the separation, the identification of the outgoing momenta with the eigenvalues of L is one line. That the N -body scattering map is a product of two-body maps is Moser's too, a statement about the asymptotic motion, not a claim that the particles meet only in pairs; here the three-body composition is checked numerically.
- *A classical counterpart, not a limit,* for the delta gas: classical point particles with a contact interaction show no displacement at all (Sutherland §4.2).
- *The linear terms of the spring* are dropped by fiat on the open chain; the end particles go to infinity, where they would be an infinite constant.
- *The open chain's own integrability* is not used here, only its Lax matrix (Lecture 8) and Moser's theorem. Its proof, and the reason its fibres are not compact, are in Lecture 9's readings.

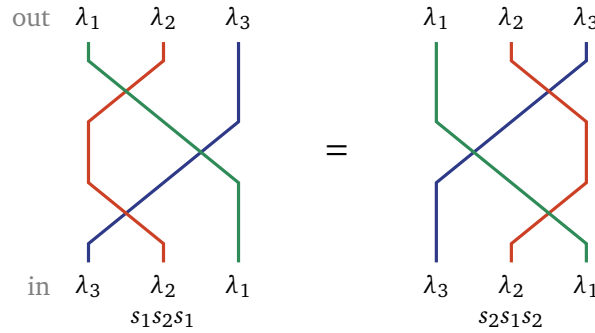


Figure 12.6: The three-body scattering of Figure 12.5 as a permutation of momentum lines, time upward, drawn schematically. Each crossing stands for a two-body collision, which exchanges two adjacent momenta and displaces both lines by the two-body amount of Proposition 12.1. The two orders are the braid relation $s_1s_2s_1 = s_2s_1s_2$, and they carry the same total displacement on each line. The same picture with lines for particles and crossings for R -matrices is the Yang–Baxter equation of Lecture 13.

Reading

12.7 FPUT, recalled

In 1955 Fermi, Pasta, Ulam and Tsingou put a chain of masses joined by weakly nonlinear springs on a computer, with spring energy

$$V_{\text{FPUT}}(r) = \frac{r^2}{2} + \alpha \frac{r^3}{3}, \quad (12.7)$$

r the extension of the spring. They started it with all the energy in the lowest normal mode and expected the nonlinearity to share it out among all the modes. It did not: the energy wandered into a few neighbouring modes and came back, almost completely (Figure 1). A system that behaves this regularly is behaving as if it had more integrals than energy.

Reading

12.8 Why the open chain's fibres are planes

Lecture 9's readings prove the open chain Liouville integrable and each connected component of its level sets a cylinder $T^k \times \mathbb{R}^{N-k}$ (Corollary 9.24), with the number k of periodic directions left open. The scattering picture settles it.

Proposition 12.3 (The fibres are planes · sketched). *In Corollary 9.24, $k = 0$: every Σ is diffeomorphic to $\mathbb{R}^N$.*

Proof (sketch). A stabiliser element $t \neq 0$ means that the flow Φ of $G = \sum_k t_k F_k$ returns every point of Σ to itself at time 1; in particular it returns the points $x(t)$ of a trajectory at late times. There, by Theorem 12.2, the particles are far apart and $L \simeq \text{diag}(\lambda)$. By (9.20) X_G then moves particle n at the speed $\partial G / \partial p_n \simeq g(\lambda_n)$, $g(\xi) = \sum_k k t_k \xi^{k-1}$; its speed is bounded on Σ (proof of Proposition 9.23), so in unit time the particles stay far apart and the momenta

barely change. Returning in time 1 forces $g(\lambda_n) = 0$ for every n , hence $g = 0$ and $t = 0$ by the Vandermonde argument of [Theorem 9.20](#). The stabiliser is trivial. ■

On a torus the commuting flows have a full lattice of periods, even though a single irrational flow need never return ([Figure 5.2](#)). On the open chain's fibre no combination of the flows ever returns; the proof used the physical flow of H , whose late-time configurations are free particles moving apart. In effect $\Sigma \cong \mathbb{R}^N$ is coordinatised by the incoming offsets β_n^- of [Theorem 12.2](#), while the asymptotic momenta are fixed on it ([Figure 12.7](#)). They are the integrals, and they are also the obstruction: as long as they are distinct, the particles separate.

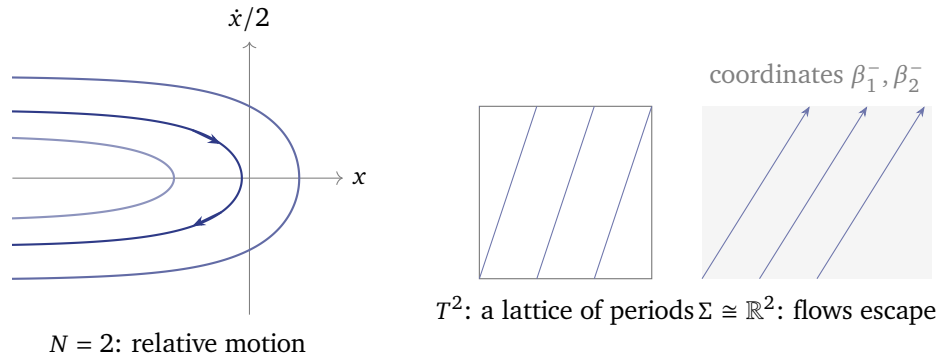


Figure 12.7: Non-compact fibres. Left: for $N = 2$ the relative motion lies on the curves $\frac{1}{4}\dot{x}^2 + e^x = v^2/4$ of (12.3), three of which are drawn. Each is a line open to $x \rightarrow -\infty$, run through once: the particles approach, turn and separate. Together with the free centre-of-mass line this makes each fibre a plane. Right: the contrast with [Theorem 5.7](#). On a torus the commuting flows wrap round, with a full lattice of periods; on the open chain's fibre they are straight lines in the incoming offsets and never return ([Proposition 12.3](#)).

Close the chain into a ring and this changes. The whole ring can still drift (the flow of F_1 is still a translation), but the particles can no longer separate. Once the total momentum is fixed and the common translation split off, the relative level sets are compact, and each regular connected component is a torus T^{N-1} , whose actions are read off a curve. That is [Lecture 9](#).

Summary.

- *The model:* the Toda spring $e^{-r} + r - 1$ is FPUT's to cubic order; the open chain (12.2) is the Toda ring without its closing spring.
- *Two particles* ([Proposition 12.1](#)): the momenta are exchanged, they are the eigenvalues of L at every time, and each momentum line is shifted by $\mp 2 \ln v$.
- *N particles* ([Theorem 12.2](#), cited): the outgoing momenta are the incoming ones, permuted, and the shifts are sums of two-body shifts (12.6): elastic, non-diffractive scattering.
- *Three bodies:* the two ways of composing the pairwise scatterings give the same shifts, the braid relation of [Figure 12.6](#).
- *Next:* [Lecture 13](#) asks what makes factorised scattering consistent when the particles carry a label, and finds the Yang–Baxter equation.

Tutorial. Exercise Sheet 12.

References

J. Moser, in *Dynamical Systems, Theory and Applications*, Lecture Notes in Physics 38 (1975) 467; Toda, *Theory of Nonlinear Lattices*, §§2.1, 3.10; Babelon, Bernard and Talon, *Introduction to Classical Integrable Systems*, §4.8; Sutherland, *Beautiful Models*, §4.2.